

Dwarf planets

In 1930, a new planet was found beyond the orbit of Neptune. The new body was named Pluto, and it was found to be by far the smallest planet, even smaller than Earth's Moon. By 2005, improved survey techniques had found several bodies similar in size to Pluto in the same area of the Solar System, and one (Ceres) in the Main Belt of asteroids. It was then decided to name objects of this size range dwarf planets, and Pluto became one of them.

▷ Independent bodies

Dwarf planets are independent bodies large enough to have become almost spherical because of their internal gravity, but are too small to be called planets.

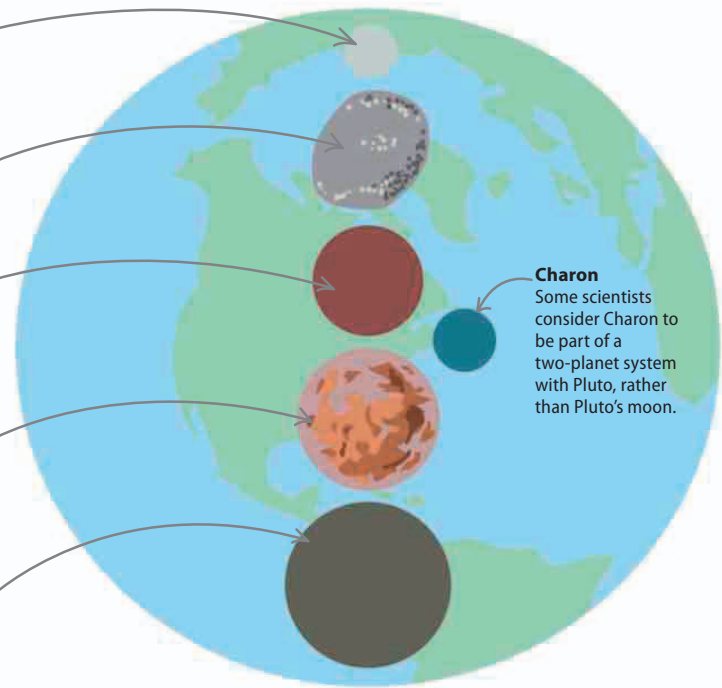
Ceres
This is the largest body in the Main Belt of asteroids. It is made of rock and metal.

Haumea
This misshapen body in the Kuiper Belt is largely made of ice.

Makemake
A very cold ball of ice, this dwarf planet comprises frozen methane, ammonia, and water.

Pluto
Pluto has five moons. The largest is Charon, about half the size of Pluto.

Eris
Discovered in 2005, Eris is currently the largest dwarf planet known.



Comets

Comets are "dirty snowballs" of dust and ice formed at the birth of the Solar System. They have highly elliptical (oval-shaped) orbits. Some travel far beyond Neptune, taking thousands of years to circle the Sun, while others have short paths of just a few years. As a comet nears the Sun it heats up and dust and gas stream out, forming tails millions of kilometers long.



△ Comet tails

When a comet gets closer to the Sun, beyond the orbit of Mars, it becomes active, developing a coma and tails. When the comet travels out of the inner Solar System these disappear again.

REAL WORLD

Meteoroids and meteorites

Meteoroids are chunks of space rock that produce streaks of light, called meteors, as they burn up in Earth's atmosphere. Most are small and around 3,000 tons of space rock hits Earth every year as dust. A meteoroid that survives the atmosphere to land on Earth's surface is called a meteorite and these can form large impact craters. Roter Kamm in Namibia, for example, is more than five million years old, 2.5 km (1.5 miles) wide, and 130 m (425 ft) deep.

